

Neel Gajare

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. EECS (Electrical Engineering & Computer Science)

Dec 2026

Relevant Coursework (3.9 GPA): Scalable AI, Advanced Topics in Computer Systems, Advanced Algorithms, Computer Architecture, Machine Learning, Operating Systems, Computer Security, Database Management, Quantum Computing, Optimization Models in Eng, Circuit Design, Discrete Math & Probability, Electricity & Magnetism, Mechanics, Competitive Programming, Digital Design and Integrated Circuits, FPGA Lab, Advanced LLM Agents
Societies: Space Technology @ Cal, Cloud Computing @ Cal, Codify, Eta Kappa Nu (EECS Honor Society), Sky Computing Lab

EXPERIENCE

ML Intern @ Apple

May 2026 – Aug 2026

Apple

Santa Clara, CA

- Worked on developing agentic AI systems and training neural network models such as GNNs to accelerate and optimize physical design workflows such as placement & routing and timing path diagnosis for next-generation Apple silicon.

AI Researcher

Feb 2024 – Present

Berkeley AI Research Lab (BAIR) & Sky Computing Lab

Berkeley, CA

- Sky Computing Lab: Working under Professor Alvin Cheung on LLM-driven kernel optimizations on tensor accelerators which utilizes a beam search over LLM-generated optimization plans, pruned by correctness tests and cycle-accurate performance feedback. Achieved $1.4\times$ to $3\times$ performance over various expert hand-tuned schedules on TPU v6. Previously worked on using thought trees to predict reasoning models accuracy on coding tasks. Paper published in COLM.
- BAIR: Worked under Professor Kurt Keutzer on Generative AI project pertaining to low resource machine translation and linguistic segmentation. Utilized Langchain, Chroma, Milvus to implement efficient RAG techniques on billion-token corpus to improve translational model. Integrated vector database to backend system and API with FastAPI. Implemented agentic workflows for paragraph segmentation techniques to enhance retrieval system.

SWE Intern @ Amazon

May 2025 – Aug 2025

Amazon

Sunnyvale, CA

- Designed low-latency traffic analysis workflows to detect and identify unique non-volumetric and agentic bot signatures affecting Amazon's retail websites and backend services.
- Utilized AWS tools such as Athena, Sagemaker AI, Cloudwatch, EC2, and internal tools to automate manual traffic spike investigation time for oncall engineers by $>90\%$ and reduce compute costs by $>85\%$ on average.

Stanford Research Intern

May 2022 – Feb 2024

Stanford University

Palo Alto, CA

- Worked in Stanford's Crustal Deformation Lab under Professor Paul Segall. Researched complex basaltic magma chamber geometries using AI.
- 1st Author of abstract published in American Geophysical Union (AGU) conference proceedings, presented in AGU Fall Meeting. Utilized Gmsh for 3-D mesh generation to model magma chambers, Matlab for simulation of surface deformation and Bayesian optimization techniques.

AWARDS

- Apple FPGA Digital Design Contest - 1st Place May 2025
- Nvidia National Merit Scholarship April 2023
- St. Francis de Assisi Volunteering Award May 2022

PROJECTS

RISC-V CPU on FPGA | Verilog

Jan 2025 – May 2025

- Implemented 5-stage pipelined RISC-V CPU on FPGA, with UART tethering, BIOS functionality, synchronous memories.
- Optimized CPU to run on 125 MHz clock frequency and 1.06 average CPI on various benchmarks by resolving pipelining hazards, designing a pipelined hybrid Gshare branch predictor + branch target buffer + return address stack.
- Achieved highest figure-of-merit score in Berkeley's digital design competition sponsored by Apple.

IntrospectAI | React, Three.js, Flask, Claude, Supabase

Jun 2025

- Built real-time 3D visualization tool that transforms ChatGPT and Claude conversation histories into interactive, AI-labeled semantic clusters using Voyage embeddings and Claude Sonnet 4 for topic detection and RAG.
- Created a high-performance React + Three.js frontend with GPU-accelerated WebGL shaders and a Flask backend supporting multi-user interaction, live API updates, semantic search (pgvector), and dynamic clustering.
- Designed a resilient NLP pipeline with SpaCy, UMAP, KMeans, and Claude for auto-labeling; supported cross-platform chat parsing, mock data fallback, and smooth UX for filtering, zooming, and exploration of AI chat patterns.

TECHNICAL SKILLS

Languages: Python, C, C++, Verilog, SQL, RISC-V Assembly, JavaScript

Frameworks/Tools: vLLM, Triton, LangGraph, AWS EC2, Athena, Sagemaker AI, Docker, Kubernetes, Trainium, LangChain, Pytorch, Vivado, Keras, OpenCV, Gmsh, Numpy, Jax, MongoDB, OpenAI, HuggingFace, Chroma, FastAPI, Firebase, Valgrind, Streamlit, Milvus, GDB, TensorRT, PrimeTime, NSight Systems/Compute, SGLang, CUDA, Pallas

Skills: AI/ML, Chip Design, Computer Architecture, Algorithms, Data Processing/Analysis, EDA, NLP